A local replacement for the filtration argument in GTZ Lemma 13.2

Project proof draft: /root/astra_v14_baseline_theta

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Status. This is a proposed project proof, not an independent acceptance. It replaces only the multilinearity lemma and its auxiliary filtration construction. It does not certify the other corrections to the inverse theorem. All source and consumer bindings are recorded separately.

1 Statement and conventions

We use the nilmanifold, polynomial-map and nilcharacter conventions of [1, Sections 3, 4 and 6]. In particular all Lie groups below are connected, simply connected and nilpotent; filtration groups are connected rational Lie subgroups. A rational subgroup J means that $J \cap \Gamma$ is a lattice in J. The multidegree ordering is coordinatewise. Write $\mathbf{1} = (1, \dots, 1) \in \mathbb{N}^s$ and $|a| = \sum_j a_j$. The group G_a is trivial when $a \not\leq \mathbf{1}$. We use $\mathrm{e}(t) = \exp(2\pi i t)$ and complex conjugation componentwise in tensor products. A vector-valued nilcharacter has a unit-vector-valued Lipschitz test function and a real vertical character, as in [1, Definition 6.1]. For limit objects, every defining relation is read in its natural ultrapower extension, using [1, Appendix A, Definition A.2 and the following discussion, and Definition A.3, pp. 1324–1325]. In particular the vertical identity is required for every limit point $x \in {}^*(G/\Gamma)$ and every $t \in {}^*G_{\mathbf{1}}$, with the standard homomorphism η extended to ${}^*\eta$. This is the internal nilcharacter convention used in this proof. We do not infer this exact identity from an identity only at standard points; bounded Lipschitz continuity alone would not justify that inference.

Theorem 1 (The unchanged multilinearity statement)**.** *Let $s, m \geq 1$ be fixed standard integers. Let χ be a nilcharacter of multidegree at most $\mathbf{1}$ on $\mathbb{Z}^s$, or a limit nilcharacter with that description in the sense of [1, Sections 5–6]. For fixed homomorphisms $L_1, L_1', L_2, \dots, L_s : \mathbb{Z}^m \longrightarrow \mathbb{Z}$, the three pullbacks below are nilcharacters of degree at most s and satisfy*

$$[\chi(L_1 + L_1', L_2, \dots, L_s)]_{\mathrm{Symb}^s} = [\chi(L_1, L_2, \dots, L_s)]_{\mathrm{Symb}^s} + [\chi(L_1', L_2, \dots, L_s)]_{\mathrm{Symb}^s}.$$

The same assertion holds in each coordinate. More explicitly, each component of

$$\chi(x + x', y_2, \dots, y_s) \otimes \overline{\chi(x, y_2, \dots, y_s)} \otimes \overline{\chi(x', y_2, \dots, y_s)} \tag{1}$$

is a polynomial nilsequence of degree at most $s-1$ on $\mathbb{Z}^{s+1}$. The representing nilmanifold is fixed by the original filtered nilmanifold; its test-function Lipschitz bound is bounded in terms of that fixed data and the original Lipschitz bound, independently of the polynomial coefficients.

The symbol notation abbreviates the degree-s equivalence relation of [1, Definitions 6.6 and 6.22]. Thus the explicit assertion about (1) supplies the asserted equality, with no approximation or loss in degree. It is enough to prove the ordinary statement with uniform fixed-data bounds and then take limit sequences. We supply that uniformity below. We do not assume G_0 equals the group generated by its positive multidegree groups.

2 Subgroups and lattices used in the construction

We use the following basic nilpotent Lie-group facts recorded in [1, Section 3, pp. 1242–1243]: the exponential is a homeomorphism; the connected subgroup generated by a set is the exponential of the generated Lie algebra; and, for a finite set A, its discrete generated group is cocompact in its connected real generated group. The last assertion is applied only when A lies in an already specified discrete lattice. Here are the compactness details needed to apply these facts.

Lemma 2. *Let M be a connected simply connected nilpotent Lie group with lattice Λ. Let $J_1, \ldots, J_t$ be connected rational subgroups. Then the connected subgroup generated by the J_i is closed, simply connected, and rational in M. This also applies to diagonal or coordinate embeddings of such subgroups into a finite product of M.*

Proof. First a lattice Λ_J in a connected Lie group J is finitely generated. Choose a proper left-invariant path metric, and R such that every point is within distance R of some lattice point; compactness of J/Λ_J, followed by inversion, gives such an R. Follow a path from the identity to a lattice point in steps of length at most one and choose nearby lattice points, using the prescribed endpoints. Consecutive chosen lattice points differ by an element in the compact ball of radius $2R + 1$. Its intersection with the discrete lattice is finite and generates the lattice.

The connected real subgroup generated by this finite generating set is all of J. To justify this last assertion, let U be that subgroup. It contains Λ_J, so J/U is compact. A connected subgroup of a simply connected nilpotent group is closed, and its homogeneous space is a Euclidean space. One can see the latter directly by a chain of Lie subalgebras from $\mathfrak{u}$ to $\mathfrak{j}$, each of codimension one and an ideal in the next: the normalizer of a proper subalgebra of a nilpotent Lie algebra strictly contains it, so such a chain is obtained by choosing one normalizing vector at each step. Successive exponential coordinates give the Euclidean coordinates on J/U. For completeness, the normalizer assertion follows by taking the first member of the ascending central series not contained in the subalgebra; the preceding member is contained in it, so any vector in that member but outside the subalgebra normalizes it. Hence compactness forces $U = J$.

Choose a finite generating set for each $J_i \cap \Lambda$. Their union A lies in Λ and generates all the J_i as a connected real group. Its abstract generated group is discrete because it lies in Λ. The finite-set fact from [1, p. 1243] therefore makes it a lattice in $\langle A \rangle_{\mathbb{R}}$. Consequently the larger group $\Lambda \cap \langle A \rangle_{\mathbb{R}}$ is also a lattice there. A Lie subalgebra of a nilpotent Lie algebra exponentiates to a closed simply connected subgroup, which proves the other assertions. For a diagonal or coordinate embedding, apply the same argument to the images of the finite generating sets, which lie in the product lattice. □

We will also use the elementary central-quotient consequence explicitly. If J is such a group, Δ its lattice, and C a connected central rational subgroup, then J/C is connected simply connected nilpotent and the image of Δ is a lattice. Cocompactness follows from the continuous image of J/Δ. For discreteness, if lattice cosets tend to the identity coset, choose local lifts tending to the identity and write the lattice points as those lifts times elements of C. Reduce those elements modulo the lattice $C \cap \Delta$ into a fixed compact set. The adjusted lattice points now lie in a compact subset of J, so only finitely many occur. Thus their cosets cannot accumulate at the identity. Simple connectedness follows either from exponential coordinates or the homotopy sequence with connected kernel C.

3 Bounded normalization at zero

Write $\chi(n) = F(g(n)\Gamma)$, with the data in Theorem 1. Initially restrict to the fixed subgroup G_0 and its lattice $\Gamma_0 = G_0 \cap \Gamma$, restricting F through the smooth inclusion $G_0/\Gamma_0 \longrightarrow G/\Gamma$. This only changes a fixed metric comparison constant. In the following notation $G = G_0$ and $\Gamma = \Gamma_0$.

Choose a fixed compact set of representatives for G/Γ. Write $g(0) = q\gamma$, with q in that compact set and $\gamma \in \Gamma$. Set

$$h(n) = q^{-1}g(n)\gamma^{-1} = q^{-1}(g(n)g(0)^{-1})q, \qquad F_q(z\Gamma) = F(qz\Gamma). \tag{2}$$

Then $h(0) = 1$ and $F_q(h(n)\Gamma) = F(g(n)\Gamma)$ exactly. Each G_a is normal in G_0, because $[G_0, G_a] \subseteq G_a$. The constant map with value $g(0)^{-1}$ is polynomial into the original filtration. Product closure of polynomial maps [1, Corollary B.4], followed by the filtered conjugation by q, shows that h is polynomial into the same multidegree filtration. No conjugation of the lattice is made.

The maps $z\Gamma \mapsto qz\Gamma$ are smooth with uniformly bounded derivatives for q in the fixed compact set. Thus F_q has a uniform Lipschitz bound. If $F(tz) = \mathrm{e}(\eta(t))F(z)$ for $t \in G_{\mathbf{1}}$, then

$$F_q(tz) = \mathrm{e}(\eta_q(t))F_q(z), \qquad \eta_q(t) = \eta(qtq^{-1}). \tag{3}$$

This is a homomorphism on $G_{\mathbf{1}}$. Its restriction to $G_{\mathbf{1}} \cap \Gamma$ is integer-valued: put $z = \Gamma$ in (3) and use both $F_q(t\Gamma) = F_q(\Gamma)$ and the fact that $F_q(\Gamma)$ is a unit vector. In fact $\eta_q = \eta$. For each fixed $t \in G_{\mathbf{1}} \cap \Gamma$, the continuous map $q \mapsto \eta(qtq^{-1})$ on connected G_0 has integer values, so it equals its value $\eta(t)$ at the identity. The lattice generates $G_{\mathbf{1}}$ as a connected real group, by the argument in Lemma 2. Two continuous real homomorphisms agreeing on that lattice therefore agree on $G_{\mathbf{1}}$. This proves the assertion for every $q \in G_0$. In particular the normalization preserves the same standard vertical character, including for limit choices of q.

4 The descending auxiliary filtration

For $a \in \{0,1\}^s$ with $a \neq 0$, define the following homomorphisms on G_a whenever the indicated first coordinate occurs:

$$A_a(u) = (u,u,u) \quad (a_1 = 0), \qquad B_a(u) = (u,u,1), \quad C_a(u) = (u,1,u) \quad (a_1 = 1).$$

For $1 \leq i \leq s$ let H_i be the connected subgroup of G^3 generated by these images for all a with $|a| \geq i$. Set $H_{s+1} = \{1\}$ and

$$K = H_1, \qquad K_0 = K_1 = H_1, \qquad K_i = H_i \ (i \geq 1). \tag{4}$$

Every group just defined is fixed by the original filtration, not by g. Lemma 2 makes it a connected closed simply connected rational subgroup of G^3. Put $\Delta = K \cap \Gamma^3$. The sequence (K_i) is descending by its definition.

We check the commutator law. For generators of types A, A, A, B, and A, C, the coordinatewise commutator has respectively type A, B, C and belongs to the group with index $a + b$. The weight adds. If two generators both have first index one, every possibly nontrivial coordinate commutator belongs to $G_{a+b} = \{1\}$ because $(a+b)_1 = 2$. Thus the B, B, B, C, and C, C commutators are trivial. These assertions hold for their Lie algebras as well. Bracketing generated Lie algebras and using the Jacobi identity gives

$$[\mathfrak{h}_i, \mathfrak{h}_j] \subseteq \mathfrak{h}_{i+j} \quad (i, j \geq 1).$$

Here iterated brackets remain in the indicated space: each bracket is a sum of iterated brackets of the original generators, and its total original weight is at least $i + j$. BCH, which terminates for a nilpotent group, then gives $[H_i, H_j] \subseteq H_{i+j}$. The index-zero assertions also hold since $K_0 = H_1$: for example $[K_0, K_j] \subseteq H_{j+1} \subseteq H_j = K_j$. Hence (K_i) is a rational degree filtration, is taut at zero, and has degree at most s.

The top group has a particularly simple exact description:

$$K_s = H_s = \{(uv, u, v) : u, v \in G_{\mathbf{1}}\}. \tag{5}$$

Indeed $a = \mathbf{1}$ is the only index of weight s and has first coordinate one. The group $G_{\mathbf{1}}$ is abelian since $[G_{\mathbf{1}}, G_{\mathbf{1}}] \subseteq G_{\mathbf{21}} = \{1\}$. Thus its B and C images commute and their product is exactly the displayed connected group. It is central in K, also following from the commutator law.

5 The exact polynomial orbit

Choose once and for all a total order of $\{0,1\}^s$ extending its partial order. By [1, Lemma B.9], the normalized map has the Taylor expansion

$$h(n) = \prod_{a \in \{0,1\}^s \setminus \{0\}} u_a^{n^a}, \qquad u_a \in G_a, \qquad n^a = \prod_{j=1}^{s} n_j^{a_j}. \tag{6}$$

The absent constant coefficient equals $h(0) = 1$. Define

$$P(x, x', y) = (h(x + x', y), h(x, y), h(x', y)), \qquad y = (y_2, \ldots, y_s).$$

This is exactly the ordered product of the following factors, in the order of (6). If $a_1 = 0$, its factor is

$$A_a(u_a)^{\prod_{j=2}^{s} y_j^{a_j}}.$$

If $a_1 = 1$, its factor is

$$B_a(u_a)^{x \prod_{j=2}^{s} y_j^{a_j}} C_a(u_a)^{x' \prod_{j=2}^{s} y_j^{a_j}}.$$

The latter identity is exact because the powers of the same element u_a commute. In either case the integer exponent has degree $|a|$ and the fixed group element lies in $K_{|a|}$. A map $v^{Q(n)}$ of this form is polynomial into (K_i): its j-fold derivative is $v^{\Delta^j Q(n)}$, which lies in K_j for $j \leq \deg Q$, and is the identity for $j > \deg Q$. Here derivatives commute within the cyclic one-parameter subgroup generated by v, and $K_{\deg Q} \subseteq K_j$ is used in the first case. The values also lie in K_0. Product closure [1, Corollary B.4] now proves that

$$P \in \mathrm{poly}(\mathbb{Z}_{\mathbb{N}}^{s+1} \longrightarrow K_{\mathbb{N}}). \tag{7}$$

This argument proves the assertion for all integer inputs, not just a box. In particular it verifies that the orbit lies in the normalized ambient group K, which the exact-weight construction at index zero did not ensure.

6 The lower-degree quotient and its test function

Let $\overline{K} = K/K_s$, let $\pi : K \longrightarrow \overline{K}$ be the quotient map, and let $\overline{\Delta} = \pi(\Delta)$. The central-quotient argument after Lemma 2 shows that $\overline{K}/\overline{\Delta}$ is a compact nilmanifold with connected simply connected

nilpotent group. Give it the filtration $\overline{K}_i = \pi(K_i)$. It is taut and has degree at most $s-1$. For each $i \le s$, $K_s \subseteq K_i$, so

$$\overline{\Delta} \cap \pi(K_i) = \pi(\Delta \cap K_i).$$

The right side is cocompact in $\pi(K_i)$, by the compact quotient of $K_i/(\Delta \cap K_i)$; it is discrete as a subgroup of $\overline{\Delta}$. Thus the quotient filtration is rational at every index. Equation (7) remains polynomial after applying π, directly from the derivative definition.

Define on K/Δ

$$T_q(z_1, z_2, z_3) = F_q(z_1\Gamma) \otimes \overline{F_q(z_2\Gamma)} \otimes \overline{F_q(z_3\Gamma)}.$$

It is well-defined and Lipschitz, by restriction from $(G/\Gamma)^3$. For $(uv, u, v) \in K_s$, equation (3) multiplies this tensor by

$$\mathrm{e}\big(\eta_q(uv) - \eta_q(u) - \eta_q(v)\big) = 1.$$

Therefore T_q is invariant under K_s and descends to a function $\overline{T}_q$ on $\overline{K}/\overline{\Delta}$.

For explicit control of its regularity, fix a smooth metric on K/Δ and average it over the compact central torus $K_s/(K_s \cap \Delta)$. The averaged metric and the original fixed metric are uniformly equivalent. The quotient metric makes the quotient a Riemannian submersion. A function constant on its fibers and Lipschitz upstairs is Lipschitz downstairs with the same bound for these metrics, by horizontal path lifting. All the metric comparisons are fixed by the original filtered nilmanifold. The bound for T_q is at most a fixed multiple of the bound for F_q, since its three factors have norm one. The earlier compact normalization therefore makes the Lipschitz bound for $\overline{T}_q$ uniform in g.

Finally the expression (1) is exactly

$$\overline{T}_q(\pi(P(x, x', y))\overline{\Delta}).$$

This proves the claimed degree at most $s-1$ representation.

For clarity, the individual factors also have valid degree-s representations. In one copy of G, take the connected group J_i generated by G_a for $|a| \ge i \ge 1$, and set $J_0 = J_1$. The same rationality and commutator arguments give a taut rational degree-s filtration on J_1, with $J_s = G_{\mathbf{1}}$. The Taylor factors of h and of each of its linear pullbacks are polynomial for this filtration by the same integer-polynomial argument. Restricting F_q gives a unit-vector-valued Lipschitz test function with vertical character η_q. Hence those factors are nilcharacters of degree at most s.

Substitute the homomorphism $n \mapsto (L_1(n), L_1'(n), L_2(n), \dots, L_s(n))$ into the lower-degree representation. Derivatives, or the explicit polynomial factors above, show that degree does not increase. The definition of symbol equivalence proves Theorem 1. Relabeling coordinates gives the other coordinate identities.

7 Limit data and the actual published consumers

For a limit nilcharacter the filtered nilmanifold and its metric data are standard fixed data, and the polynomial and Lipschitz test functions are limit sequences with bounded Lipschitz norm. Choose the bounded representatives q separately in each member of those sequences; this gives a limit map with values in the same compact set. All groups K_i, the lattice, the quotient and the metric constants constructed above are unchanged along the sequence. The same standard vertical character is retained by the preceding argument. Representatives for the test functions may be taken to satisfy their defining unit-norm and vertical-character identities on an ultrafilter-large set of indices: otherwise, choosing a violating point, and for the second identity a violating group element, at every bad index would contradict the exact identity of the limit function at the

resulting limit points. This uses the all-limit quantifiers just specified and the evaluation rule in [1, Definition A.3]; it would be invalid with only standard-point quantifiers. Discard those bad indices. The formulas now hold at every remaining member, and the test-function bounds are uniform. Their limit is therefore a limit polynomial nilsequence of degree at most $s-1$ in precisely the required sense. No compactness argument selects a nilmanifold depending on an unbounded polynomial coefficient.

The first published use, immediately after Lemma 13.2 on p. 1318, expands $[\widetilde{\chi}(n+h,\dots,n+h)]$ by applying the identity in each coordinate. Before any use of symmetry the exact expansion is

$$[\widetilde{\chi}(n+h,\dots,n+h)] = \sum_{\omega\in\{0,1\}^s} [\widetilde{\chi}(z_{\omega_1},\dots,z_{\omega_s})], \qquad z_0 = n,\ z_1 = h.$$

Every map used here is a fixed integer homomorphism from $\mathbb{Z}^2$. Thus the coefficient counting and subsequent symmetry step in that argument receive exactly the same multilinearity statement; no stronger inverse theorem is asserted by this local repair.

The further explicit appeals in Lemmas 13.3, 13.4 and 13.5 on pp. 1319–1322 likewise use addition, subtraction and repeated copies of their variables. These are integer homomorphisms from the displayed finite Cartesian powers of $\mathbb{Z}$, so the theorem applies without a range or normalization change. Constant shifts in those arguments still require the separate symbol-shift assertion in E.8; this proof does not discharge it. The repaired lemma supplies the multilinearity input to Theorem 7.4, then to the published inverse-theorem argument. Other corrections, including the false original Proposition 8.3 and its replacement, remain separate obligations. Source and current-manuscript consumer spans are recorded in the accompanying contracts.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics **176** (2012), 1231–1372. The frozen final published PDF is the source for the numbered results used here. The arXiv v5 correction motivates the auxiliary generator types but is not used as an accepted theorem in this proof.